KS3 DIGITAL TECHNOLOGY

DATA SCIENCE AND DATA VISUALISATION

Turn raw data into answers: ask good questions, collect and clean data, find patterns with averages and correlation, choose honest charts and tell a clear story with data.

LESSONS

1 What is data science?
2 Collecting data
3 Cleaning data
4 Finding patterns
5 Charts that tell the truth
6 Data stories and privacy

LESSON 1 OF 6

WHAT IS DATA SCIENCE?

Data science is using data to answer questions and make decisions.

STARTER · 5 MINUTES

DO NOW

Your phone knows how many steps you walked every day this year. What questions could you answer with that data?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe what data science is and the stages of the data science cycle

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what data science is
- ✓ I can name the six stages of the data science cycle
- ✓ I can give examples of data science in real life

QUESTIONS ANSWERED WITH DATA

- Data science means using data to answer questions and make better decisions
- Which bus routes need more buses at 8am?
- Which players should a football club sign?
- When will A&E be busiest this winter?

KEY WORDS

DATA SCIENCE	Using data to answer questions, find patterns and make decisions.
DATA	Raw facts and figures, such as numbers, words or measurements, before they are given meaning.
BIG DATA	Datasets so large, fast or varied that ordinary tools can't handle them.

THE DATA SCIENCE CYCLE

- 1 Ask a clear question
- 2 Collect the data
- 3 Clean it
- 4 Analyse it to find patterns
- 5 Visualise it with charts
- 6 Communicate the answer, then ask the next question

ACTIVITY · 15 MINUTES · PAIRS

SHARPEN THE QUESTION

1. Read the five vague questions on the board (e.g. "Is the canteen good?").
2. Rewrite each as a specific, measurable question that data could answer.
3. For one of them, list what data you would need to collect.

In your workbook, answer:

? Write your improved version of "Is the canteen good?" and the data you'd need.

? Why is a specific question better for data science?

Your workbook: Lesson 1 · tick each step off as you go

BIG DATA

- Some datasets are enormous: billions of rows, arriving every second
- Volume (how much), velocity (how fast), variety (how many kinds)
- Examples: card payments, weather sensors, streaming services, smart meters
- Needs powerful computers and clever programs to analyse

QUICK CHECK

1. What is data science?

A Using data to answer questions and make decisions
B Typing data quickly
C Writing science fiction
D Building computers

2. What is the first stage of the data science cycle?

A Ask a clear question
B Clean the data
C Buy a computer
D Make a chart

3. What is big data?

A A data cable
B A large font
C A big spreadsheet cell
D Datasets so large or fast that ordinary tools can't handle them

4. Which "V" of big data means how fast data arrives?

A Variety
B Value
C Velocity
D Volume

QUICK CHECK: ANSWERS

1. **A: Using data to answer questions and make decisions**
   It turns data into answers.

2. **A: Ask a clear question**
   Start with the question.

3. **D: Datasets so large or fast that ordinary tools can't handle them**
   Big data is huge, fast, varied data.

4. **C: Velocity**
   Velocity is speed.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Name the six stages of the data science cycle in order.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what data science is
- ✓ I can name the six stages of the data science cycle
- ✓ I can give examples of data science in real life

LESSON 2 OF 6

COLLECTING DATA

Data can be qualitative or quantitative, and collected first-hand (primary) or from others (secondary), including free open data.

STARTER · 5 MINUTES

DO NOW

You want to know the most popular sport in school, but you only ask the rugby team. What's wrong with your data?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Collect data fairly and describe different types and sources of data

SUCCESS CRITERIA: I CAN...

- ✓ I can tell the difference between qualitative and quantitative data
- ✓ I can compare primary and secondary data and use open data
- ✓ I can explain how sampling can make data biased

TYPES OF DATA

- Quantitative: numbers you can measure or count (height, score, minutes)
- Qualitative: descriptions, categories or opinions ("great", "blue", "rugby")
- Many datasets have both
- The type decides which charts and calculations you can use

KEY WORDS

QUANTITATIVE DATA	Data in the form of numbers that can be measured or counted.
QUALITATIVE DATA	Data in the form of words, categories, descriptions or opinions.
PRIMARY DATA	Data you collect yourself.
SECONDARY DATA	Data that has already been collected by someone else.
OPEN DATA	Data that anyone is allowed to use for free.

WHERE DOES DATA COME FROM?

- Primary data: you collect it yourself (survey, experiment, sensor)
- Secondary data: someone else collected it (websites, government statistics)
- Open data: free for anyone to use, e.g. OpenDataNI, data.gov.uk
- Always check who collected secondary data, when and how

FAIR SAMPLES

- A sample is the group you collect data from
- Biased sample: only asking the rugby team about favourite sports
- Fair samples are big enough and represent everyone
- Survey questions must be clear and not lead people to an answer

ACTIVITY · 25 MINUTES · GROUPS OF 3

SURVEY DESIGN

1. Choose a question about school life (e.g. "How do pupils travel to school, and how long does it take?").

2. Write a 5-question survey with at least one quantitative and one qualitative question.

3. Swap with another group. Find any leading, unclear or missing options in their survey.

4. Plan your sample: who will you ask, and how many?

In your workbook, answer:

? Write your two best survey questions and say whether each collects quantitative or qualitative data.

? Describe your sample and explain why it is fair.

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. Which is quantitative data?

A Type of pet
B Number of minutes spent reading
C "The film was exciting"
D Favourite colour

2. What is primary data?

A Data you collect yourself
B The first row of a table
C Data from a website
D Data about primary schools

3. What is open data?

A Data with no password
B Data that is always correct
C Data anyone is allowed to use for free
D Data about the outdoors

4. Asking only the chess club about the most popular hobby gives…

A a fair sample
B a biased sample
C big data
D open data

QUICK CHECK: ANSWERS

1. B: Number of minutes spent reading

Quantitative data is numerical.

2. A: Data you collect yourself

Primary data is first-hand.

3. C: Data anyone is allowed to use for free

Open data is free to use.

4. B: a biased sample

The sample doesn't represent everyone.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between primary and secondary data, with an example of each.

Look back at today's success criteria: can you tick them all?

- ✓ I can tell the difference between qualitative and quantitative data
- ✓ I can compare primary and secondary data and use open data
- ✓ I can explain how sampling can make data biased

LESSON 3 OF 6

CLEANING DATA

Real data is messy: typos, inconsistent formats, duplicates, missing values and impossible outliers.

STARTER · 5 MINUTES

DO NOW

In a survey, people typed their year group as "8", "Y8", "year 8", "eight" and "8A". How many different year groups would a computer think there are?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Identify and fix common problems in messy data

SUCCESS CRITERIA: I CAN...

- ✓ I can identify common problems in messy data
- ✓ I can fix inconsistent, duplicate and missing data
- ✓ I can decide what to do with outliers

DIRTY DATA

- Inconsistent formats: "8", "Y8", "Year 8"
- Typos: "Enniskilen", "Eniskillen"
- Duplicates: the same response entered twice
- Missing values: blank cells
- Impossible values: age 250, height 1.6 cm

KEY WORDS

DATA CLEANING	Finding and fixing mistakes, gaps and inconsistencies in data before analysing it.
OUTLIER	A value that is very different from the rest of the data.
DUPLICATE	An extra copy of the same record.
MISSING VALUE	A blank where data should be.

CLEANING IT

- Choose one standard format and convert everything to it
- Remove exact duplicates
- Decide how to handle blanks: remove the row, or leave it out of that calculation
- Check outliers: mistake or real?
- Keep a record of every change you make

GARBAGE IN, GARBAGE OUT

- Analysis of dirty data gives wrong answers
- Data scientists often say most of their time goes on cleaning
- Spreadsheets help: sort, filter, find and replace, remove duplicates
- Validation when collecting data stops many problems happening

ACTIVITY · 25 MINUTES · PAIRS

CLEAN THE CANTEEN SURVEY

1. Open the messy "Canteen survey" spreadsheet your teacher gives you (30 responses).
2. Find and fix: inconsistent year groups, typos in meal names, duplicates and blanks.
3. Find two outliers. Decide for each: mistake or real? Fix or keep?
4. Write a "cleaning log" listing every change you made.

In your workbook, answer:

? Write three entries from your cleaning log.

? Describe one outlier and what you decided to do with it, with a reason.

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What is data cleaning?

A Deleting all the data
B Finding and fixing problems in data before analysis
C Wiping your keyboard
D Encrypting data

2. Which is an example of inconsistent formatting?

A A blank row at the end
B Two different questions
C A chart with colours
D "Year 8", "Y8" and "8" in the same column

3. What is an outlier?

A A value very different from the rest
B The column heading
C A duplicate
D A missing value

4. What should you do with an outlier?

A Double it
B Always delete it
C Check whether it's a mistake or real before deciding
D Always keep it without checking

QUICK CHECK: ANSWERS

1. **B: Finding and fixing problems in data before analysis**
   Cleaning makes data accurate.

2. **D: "Year 8", "Y8" and "8" in the same column**
   One value written different ways.

3. **A: A value very different from the rest**
   Outliers stand out.

4. **C: Check whether it's a mistake or real before deciding**
   Investigate first.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three problems that can make data "dirty".

Look back at today's success criteria: can you tick them all?

- ✓ I can identify common problems in messy data
- ✓ I can fix inconsistent, duplicate and missing data
- ✓ I can decide what to do with outliers

LESSON 4 OF 6

FINDING PATTERNS

Averages (mean, median, mode) summarise data, and range shows spread.

DATA SCIENCE AND DATA VISUALISATION

STARTER · 5 MINUTES

DO NOW

Towns with more ice-cream sales also have more sunburn. Does ice cream cause sunburn?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Use averages, range and correlation to analyse data, and explain why correlation is not causation

SUCCESS CRITERIA: I CAN...

- ✓ I can calculate the mean, median, mode and range
- ✓ I can describe positive, negative and no correlation
- ✓ I can explain why correlation doesn't prove causation

AVERAGES AND SPREAD

- Mean: add them all up, divide by how many
- Median: the middle value when in order
- Mode: the most common value
- Range: biggest minus smallest, which shows how spread out data is

KEY WORDS

MEAN	The total of all values divided by how many there are.
MEDIAN	The middle value when data is put in order.
MODE	The value that appears most often.
RANGE	The largest value minus the smallest value.
CORRELATION	A relationship where two variables change together.

CORRELATION

- Positive: as one goes up, the other goes up (revision time and test score)
- Negative: as one goes up, the other goes down (screen time before bed and sleep)
- None: no link (shoe size and test score)
- Scatter graphs show correlation clearly

CORRELATION IS NOT CAUSATION

- Ice-cream sales and sunburn rise together, but ice cream doesn't cause sunburn
- A third factor (hot sunny weather) causes both
- Sometimes it's pure coincidence
- To show causation you need a fair experiment, not just a pattern

ACTIVITY · 25 MINUTES · PAIRS

ANALYSE THE CLASS DATA

1. Use the class dataset your teacher provides (e.g. hours of sleep, minutes of exercise and quiz scores, all anonymous).
2. Calculate the mean, median, mode and range of the quiz scores.
3. Draw a scatter graph of sleep against quiz score. Describe the correlation.
4. Suggest one other factor that might explain the pattern.

In your workbook, answer:

? Write your mean, median, mode and range, and say which average best represents the data.

? Describe the correlation and give one reason it might not mean sleep causes better scores.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is the median of 3, 7, 8, 12, 20?

A 10
B 17
C 7
D 8

2. What is the range of 4, 9, 15, 22?

A 4
B 12.5
C 18
D 22

3. Revision time and test scores both go up together. This is...

A negative correlation
B positive correlation
C causation proven
D no correlation

4. Why doesn't correlation prove causation?

A Correlation is always wrong
B Graphs can't show links
C Only experiments are interesting
D A third factor or coincidence might explain the link

QUICK CHECK: ANSWERS

1. D: 8

The middle value in order is 8.

2. C: 18

22 - 4 = 18.

3. B: positive correlation

They rise together.

4. D: A third factor or coincidence might explain the link

Patterns don't prove cause.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Calculate the mean, mode and range of: 5, 8, 8, 10, 14.

Look back at today's success criteria: can you tick them all?

- ✓ I can calculate the mean, median, mode and range
- ✓ I can describe positive, negative and no correlation
- ✓ I can explain why correlation doesn't prove causation

LESSON 5 OF 6

CHARTS THAT TELL THE TRUTH

The right chart depends on the data and the question: bar charts compare categories, line graphs show change over time, scatter graphs show relationships and pie charts show parts of a whole.

DATA SCIENCE AND DATA VISUALISATION

STARTER · 5 MINUTES

DO NOW

Look at the two charts on the board showing the same sales figures. Why does one look like a disaster and the other look fine?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Choose suitable charts for different data and identify misleading visualisations

SUCCESS CRITERIA: I CAN...

- ✓ I can choose the right chart for a dataset
- ✓ I can spot four ways charts mislead
- ✓ I can improve a misleading chart

PICK THE RIGHT CHART

- Bar chart: compare categories (favourite sport)
- Line graph: change over time (temperature each month)
- Scatter graph: relationship between two numbers (height and arm span)
- Pie chart: parts of a whole, only with a few categories

KEY WORDS

DATA VISUALISATION	Showing data as a picture, such as a chart, graph or map.
AXIS	One of the reference lines on a chart, usually horizontal (x) and vertical (y).
MISLEADING CHART	A chart that gives the wrong impression, even if its numbers are true.

HOW CHARTS LIE

- Cut-off axis: starting at 95 instead of 0 makes small differences look huge
- Cherry-picking: only showing the months that suit your story
- Wrong chart type: a pie chart that doesn't add up to 100%
- Misleading pictures: 3D effects and icons that distort sizes

ACTIVITY · 20 MINUTES · PAIRS

CHART DOCTOR

1. Your teacher gives you four "sick" charts from fictional adverts and news stories.

2. Diagnose what's wrong with each one.

3. Redraw one chart honestly (by hand or in a spreadsheet) and compare the impression it gives.

In your workbook, answer:

? For one chart, describe the trick and the impression it gives.

? Which chart type would you choose to show how a town's population changed from 1990 to 2025, and why?

Your workbook: Lesson 5 · tick each step off as you go

HONEST CHARTS

- Clear title saying what the chart shows
- Labelled axes with units
- Start bar chart axes at zero
- Show where the data came from

QUICK CHECK

1. Which chart is best for showing change over time?

A Venn diagram
B Line graph
C Pie chart
D Bar chart of categories

2. Which chart shows the relationship between two numerical variables?

A Bar chart
B Scatter graph
C Pie chart
D Pictogram

3. Why can a y-axis starting at 90 instead of 0 mislead?

A It hides the title
B It's always correct
C It makes small differences look much bigger
D It makes the chart smaller

4. What is cherry-picking?

A Picking the biggest chart
B Sorting data alphabetically
C Choosing a red colour
D Only showing data that supports your point

QUICK CHECK: ANSWERS

1. **B: Line graph**
Line graphs show trends over time.

2. **B: Scatter graph**
Scatter graphs show correlation.

3. **C: It makes small differences look much bigger**
Truncated axes exaggerate.

4. **D: Only showing data that supports your point**
It hides inconvenient data.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

State the best chart type for each: (a) favourite fruit of 30 pupils; (b) daily temperature for a month.

Look back at today's success criteria: can you tick them all?

- ✓ I can choose the right chart for a dataset
- ✓ I can spot four ways charts mislead
- ✓ I can improve a misleading chart

LESSON 6 OF 6

DATA STORIES AND PRIVACY

A data story combines a clear message, the right chart and context so people understand and act on the findings.

DATA SCIENCE AND DATA VISUALISATION

STARTER · 5 MINUTES

DO NOW

A headline says "Year 9 pupil spends 9 hours a day on their phone!" The school survey had only one Year 9 pupil from 9C. What's the problem?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Present findings as a data story and protect people's privacy when using data

SUCCESS CRITERIA: I CAN...

- ✓ I can present findings as a clear data story
- ✓ I can explain how to anonymise data
- ✓ I can describe how UK GDPR protects personal data

TELL THE STORY

- Start with your question and your headline finding
- Show the one chart that proves it best
- Give context: how much data, from whom, any limits
- Finish with what should happen next

KEY WORDS

DATA STORY	An explanation of what data shows, why it matters and what should happen next.
DASHBOARD	A screen that shows several key charts and figures together, often updated automatically.
ANONYMISE	To remove or change details so that people cannot be identified.
PERSONAL DATA	Any information that could identify a living person.
UK GDPR	The UK law, with the Data Protection Act 2018, that controls how personal data is used.

DASHBOARDS

- A dashboard shows several key charts and numbers on one screen
- Often updated live from new data
- Used by hospitals, councils, shops and sports teams
- Good dashboards show only what matters for decisions

PROTECT PEOPLE

- Personal data: anything that identifies a living person
- Anonymise: remove names and details that could identify someone
- Small groups can still identify people ("the only Year 9 in 9C")
- UK GDPR: collect only what you need, keep it secure, use it fairly

ACTIVITY · 35 MINUTES · GROUPS OF 3

DATA STORY PROJECT

1. Use your cleaned survey data (lesson 3) or an OpenDataNI dataset your teacher provides.
2. Write your question and a one-sentence headline answer.
3. Create the one chart that best supports it, with a title, labels and source.
4. Check privacy: could anyone be identified? Fix it if so.
5. Present your data story to the class in 90 seconds.

In your workbook, answer:

? Write your question, headline answer and recommendation.

? Explain one privacy risk in your data and how you dealt with it.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What is a data story?

A A clear explanation of what data shows, why it matters and what to do next
B A history of computers
C A made-up story with numbers
D A list of raw data

2. What is a dashboard?

A A screen showing key charts and figures together
B A spreadsheet formula
C A type of database
D A car's windscreen

3. What does it mean to anonymise data?

A Make the data bigger
B Delete all the data
C Encrypt the file
D Remove details that could identify people

4. Under UK GDPR, organisations should...

A keep data forever
B collect as much as possible
C collect only the personal data they need
D share data with anyone

QUICK CHECK: ANSWERS

1. **A: A clear explanation of what data shows, why it matters and what to do next**
Data stories communicate findings.

2. **A: A screen showing key charts and figures together**
Dashboards summarise data at a glance.

3. **D: Remove details that could identify people**
No one should be identifiable.

4. **C: collect only the personal data they need**
Data minimisation is a key principle.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why removing names is not always enough to anonymise data.

Look back at today's success criteria: can you tick them all?

- ✓ I can present findings as a clear data story
- ✓ I can explain how to anonymise data
- ✓ I can describe how UK GDPR protects personal data

END OF TOPIC

DATA SCIENCE AND DATA VISUALISATION

Revise each lesson in your workbook: the notes, key words and quick checks.

Then take the topic test when Mr Butler sets it.